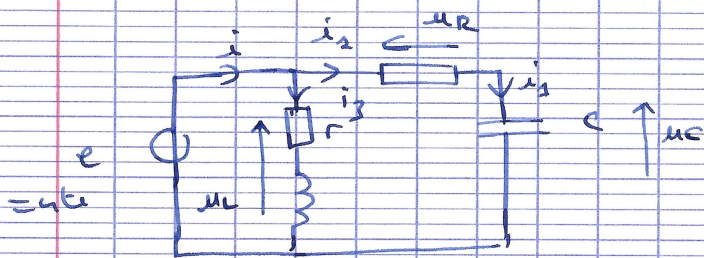


Intensité délivrée par un générateur



Branche RC :
$$e = \mu_C^{(t)} + \mu_R(t)$$

$$= \frac{q(t)}{C} + R i_2$$

$$= \frac{q_1}{C} + R \frac{dq_1}{dt}$$

$$i_2 = \frac{dq_1}{dt}$$

$$\Rightarrow \frac{dq_1}{dt} + \frac{1}{RC} q_1 = \frac{e}{R}$$

Résolv : $q_1(t) = q_1^H(t) + q_1^P$

$$q_1^P = C e$$

$$q_1(t) = A e^{-t/\tau_1} + C e$$

$$q_1^H(t) = A e^{-t/\tau_1}$$

$$\tau_1 = RC$$

à $t=0$ $q_1(0)=0 \Leftrightarrow A = -C e$

$$q_1(t) = C e \left(1 - e^{-t/\tau_1} \right)$$

Branche R, L :
$$e = \mu_L(t) = L \frac{di_2}{dt} + r i_2$$

$$= L \frac{d(i - i_1)}{dt} + r(i - i_1)$$

donc $\frac{L}{r} \frac{di}{dt} + i(t) = \frac{L}{r} \frac{di_1}{dt} + i_1 + \frac{e}{r}$

$$\frac{di}{dt} + \frac{r}{L} i(t) = \frac{di_1}{dt} + \frac{r}{L} i_1 + \frac{e}{L}$$

$$\text{Or } i_1 = \frac{dq_1}{dt} = Ce \frac{d}{dt} (1 - e^{-t/\tau_1})$$

$$= -\frac{Ce}{\tau_1} e^{-t/\tau_1} \rightarrow 0 \text{ as } t \rightarrow \infty$$

$$\frac{di_1}{dt}(t) = +\frac{Ce}{\tau_1^2} e^{-t/\tau_1}$$

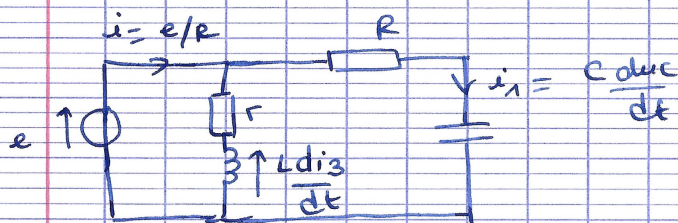
$$\frac{di(t)}{dt} + \frac{r}{L} i(t) = \frac{Ce}{\tau_1^2} e^{-t/\tau_1} + \frac{r}{L} \frac{Ce}{\tau_1} e^{-t/\tau_1} + \frac{e}{L}$$

$$i = \text{cte} \quad (\Rightarrow) \quad \forall t \quad \frac{1}{\tau_1} = \frac{r}{L} \quad \text{ie} \quad \frac{1}{RC} = \frac{r}{L}$$

On a alors

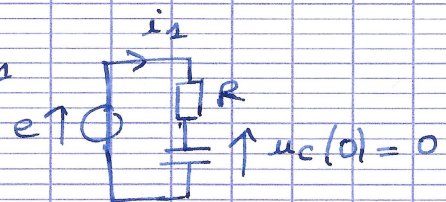
$$\text{ie } \boxed{L = rRC}$$

$$\boxed{i = \frac{e}{r}}$$



À $t=0$ $i_3 = 0$ donc $i_1 = i_2$

$$\text{d'où } i_1 = \frac{e}{R} = i(t=0)$$



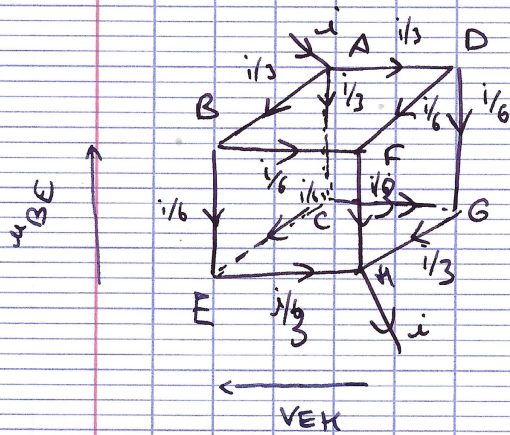
On en conclut donc que nécessairement $\boxed{r = R}$

$$\text{d'où la condition } \boxed{r = R, \quad L = rRC = R^2 C}$$

(et la conséquence $\underline{i = \frac{e}{R}}$).

Cube en fils.

①



$$i_{AD} = i_{AB} = i_{AC} = \frac{i}{3}$$

loi des noeuds en B:

$$i_{BF} = i_{BE} = \frac{i_{AB}}{2} = \frac{i}{6}$$

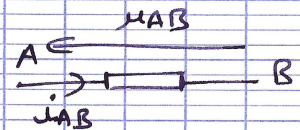
$$i_{DF} = i_{DG} = \dots$$

$$i_{FH} = i_{BF} + i_{FD}$$

$$= \frac{2i}{6} = \frac{i}{3}$$

$$i_{CE} = \frac{i}{6} = i_{CB}$$

$$i_{EH} = \frac{i}{6} \quad i_{GH} = \frac{i}{6}$$



②

$$V_A = V > 0$$

$$u_{AB}$$

$$= V_A - V_B$$

$$= r i_{AB}$$

$$= r \frac{i}{3}$$

$$V_A - V_B = r \frac{i}{3}$$

$$\Rightarrow V_B = V_A - r \frac{i}{3}$$

$$V_D = "$$

$$V_C = "$$

$$u_{BE} = V_B - V_E = r \frac{i}{6}$$

$$V_E = V_B - r \frac{i}{6} = V_A - r \frac{i}{3} - r \frac{i}{6} = V_A - \frac{3ri}{6} = V_A - \frac{ri}{2}$$

$$V_E = V_F = V_G$$

③

$$V_{EH} = V_E - V_H = r \frac{i}{3}$$

$$V_H = V_E - \frac{ri}{3} = V_A - \frac{ri}{2} - \frac{ri}{3}$$

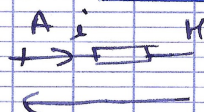
$$= V_A - \frac{3ri}{6} - \frac{2ri}{6}$$

$$= V_A - \frac{5ri}{6}$$

$$V_H = 0 \text{ donc } V_A = \frac{5ri}{6}$$

④

$$u_{AH}$$



$$u_{AH} = V_A - V_H = \frac{5ri}{6} = Ri$$

$$R = \frac{5r}{6}$$

$$R = 0,17 \Omega$$

$$⑤ \quad P = Ri^2 = \frac{5f}{6} i^2 = \frac{0,20}{6} \cdot (0,5)^2 = \underline{4,2 \cdot 10^{-2} \text{ W}}$$

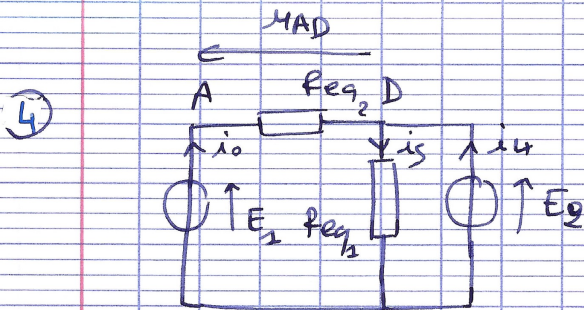
Partie 2

① B, C, D, G.

$$② \quad u = \frac{R}{3R} E_2 = \frac{1}{3} E_2 = \underline{1,0 \text{ V}}$$

$$③ \quad \frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{R} + \frac{1}{2R} = \frac{2}{2R} + \frac{2}{2R} + \frac{1}{2R} = \frac{5}{2R} \quad R_{eq} = \frac{2R}{5} = 0,40 \Omega$$

$$R_{AD} = 2R + \frac{2R}{5} = \frac{10R + 2R}{5} = \frac{12R}{5} = 2,4 \Omega$$



$$\begin{aligned} E_2 + u_{AD} &= E_1 \\ \Rightarrow u_{AD} &= E_1 - E_2 \\ &= 5,0 \text{ V} - 3,0 \text{ V} \\ &= \underline{2,0 \text{ V}} \end{aligned}$$

$$R_{eq2} i_0 = u_{AD} \Rightarrow i_0 = \frac{u_{AD}}{R_{eq2}} = \frac{u_{AD}}{R_{AD}} = \frac{5 u_{AD}}{12R} = \frac{5(E_1 - E_2)}{12R} = \underline{8,3 \cdot 10^{-1} \text{ A}}$$

⑤ On a $i_0 + i_4 = i_5$
 $\Rightarrow i_4 = i_5 - i_0$

Or $R_{eq1} i_5 = E_1$

$$i_5 = \frac{E_1}{R_{eq1}} = \frac{E_1}{3R}$$

$$i_4 = -\frac{5(E_1 - E_2)}{12R} + \frac{E_1}{3R}$$

AN $i_4 = 1,7 \cdot 10^{-1} \text{ A}$

⑥ Part diviser de courant en B:

$$i_1 = \frac{1/R}{\frac{1}{R} + \frac{1}{2R} + \frac{1}{R}} i_0$$

$$i_1 = \frac{1/R}{5/2R} i_0 = \frac{2}{5} i_0 = \frac{2}{5} \frac{(E_1 - E_2)}{12R} = \frac{1}{6} \frac{(E_1 - E_2)}{R} = 3,3 \cdot 10^{-1} \text{ A}$$

$$i_3 = i_1 = \frac{1}{6} \frac{(E_1 - E_2)}{R} \text{ par symétrie.}$$

$$i_2 = \frac{1/2R}{5/2R} i_0 = \frac{1}{5} i_0 = \frac{1}{5} \frac{(E_1 - E_2)}{12R} = \frac{E_1 - E_2}{60R} = 1,7 \cdot 10^{-1} \text{ A}$$

7

Générateur pour les 2 sources

8

$$P_{\text{refue}}(E_1) = -E_1 i_0 = -E_1 \cdot \frac{5(E_1 - E_2)}{12R}$$

$$= -4,2 \text{ W} < 0 \text{ OK.}$$

$$P_{\text{refue}}(E_2) = -E_2 i_4 = -E_2 \left[\frac{E_2}{3R} - \frac{5(E_1 - E_2)}{12R} \right]$$

$$= -0,50 \text{ W} < 0 \text{ OK}$$

$P_{\text{refue}} < 0$ comme attendu : les dipôles se comportent en générateurs.